

## HOMI BHABHA BALVAIDNYANIK COMPETITION

Class: 6 • Subject: Science • Topic: Units &amp; Measurements

# UNITS & MEASUREMENTS

## Std. 6 • Science • Homi Bhabha Examination

Focused Notes + Test-Oriented Practice

Scope: Standard units, measuring instruments, conversions, least count, estimation, measurement errors and application-based questions. Advanced topics outside the Std. 6 level are excluded.

### 1. Physical Quantities

A physical quantity is something that can be measured and expressed using a number and a unit.

Examples: Length = 5 m; Mass = 20 kg; Time = 10 s; Temperature = 30 °C.

Measurement = Numerical value + Unit

### 2. What is a Unit?

A unit is a fixed standard used for measuring a physical quantity. Without a unit, a measurement is incomplete.

### 3. Standard and SI Units

Internationally accepted standard units are used to avoid confusion. The commonly used standard system is the SI system.

| Quantity    | SI Unit  | Symbol |
|-------------|----------|--------|
| Length      | metre    | m      |
| Mass        | kilogram | kg     |
| Time        | second   | s      |
| Temperature | kelvin   | K      |

For school-level temperature questions, degree Celsius (°C) is commonly used.

### 4. Length

Length is the distance between two points.

| Conversion | Value   |
|------------|---------|
| 1 km       | 1000 m  |
| 1 m        | 100 cm  |
| 1 cm       | 10 mm   |
| 1 m        | 1000 mm |

km → m → cm → mm: towards smaller units multiply; towards larger units divide.

Examples: 4 m = 400 cm; 350 cm = 3.5 m; 2.5 km = 2500 m.

### 5. Measuring Length

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| Instrument     | Typical use                                                  |
|----------------|--------------------------------------------------------------|
| Ruler / scale  | Pencils, books, small straight objects and lines             |
| Measuring tape | Height, body measurements, cloth and larger/curved distances |

The least count of an ordinary ruler is usually  $1 \text{ mm} = 0.1 \text{ cm}$ .

- Keep the scale straight and the object along it.
- Start from zero when possible.
- Keep your eye vertically above the marking.
- Read carefully.

Damaged zero: Length = final reading – initial reading. Example:  $8 - 2 = 6 \text{ cm}$ .

## 6. Least Count

The least count is the smallest measurement an instrument can measure reliably. If the smallest division is 1 mm, least count = 1 mm.

A smaller least count generally allows more precise measurement.

## 7. Mass

Mass is the amount of matter present in an object. Its SI unit is kg.

| Conversion | Value        |
|------------|--------------|
| 1 kg       | 1000 g       |
| 1 g        | 1000 mg      |
| 1 kg       | 1,000,000 mg |

Mass is measured using a balance. Mass is amount of matter (kg); weight is gravitational force (N).

## 8. Time

The SI unit of time is second (s).

| Conversion | Value        |
|------------|--------------|
| 1 minute   | 60 seconds   |
| 1 hour     | 60 minutes   |
| 1 hour     | 3600 seconds |
| 1 day      | 24 hours     |

A stopwatch is useful for measuring short time intervals.

## 9. Temperature

Temperature tells us how hot or cold an object is. Common units include  $^{\circ}\text{C}$ , K and  $^{\circ}\text{F}$ . Temperature is measured using a thermometer.

## 10. Estimation

An estimate is an approximate measurement made when an exact value is unnecessary.

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| Object / quantity | Reasonable estimate |
|-------------------|---------------------|
| Pencil length     | About 15–20 cm      |
| Door height       | About 2 m           |
| Classroom length  | Several metres      |

Homi Bhabha questions may ask for the most reasonable measurement.

## 11. Choosing the Correct Unit

| Quantity                | Suitable unit |
|-------------------------|---------------|
| Length of a pencil      | cm            |
| Distance between cities | km            |
| Thickness of a coin     | mm            |
| Height of a person      | cm or m       |
| Mass of a school bag    | kg            |
| Mass of a small object  | g             |
| Amount of medicine      | mg            |
| Time in a race          | s             |
| School period           | min           |

## 12. Comparing Measurements

Before comparing two measurements, express them in the same unit. Example: 2 m = 200 cm, so 2 m > 150 cm.

## 13. Mixed Unit Problems

3 m 25 cm: 3 m = 300 cm, total = 325 cm. 2 kg 250 g: total = 2250 g.

## 14. Measurement Errors

Errors may arise from incorrect positioning, damaged zero mark, reading from an angle, poor calibration or human error.

Parallax error: Keep the eye vertically above the marking being read.

## 15. Homi Bhabha Quick Revision

- Measurement = numerical value + unit.
- Standard units avoid confusion.
- Smaller unit → larger numerical value for the same quantity.
- Convert to the same unit before comparing.
- Damaged zero: final – initial reading.
- Smaller least count generally means greater precision.
- Choose units according to the size/nature of the quantity.

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- Mass and weight are different quantities.

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**TEST-STYLE PRACTICE**

Instructions: Attempt without referring to the notes. Show conversion steps for numerical questions.

**A. Multiple Choice Questions**

1. Which is the SI unit of length?  
A. cm B. km C. m D. mm
2. The most suitable unit for distance between two cities is:  
A. mm B. cm C. m D. km
3. 3.5 m is equal to:  
A. 35 cm B. 350 cm C. 3500 cm D. 0.35 cm
4. Which instrument measures an athlete's race time?  
A. Ruler B. Stopwatch C. Balance D. Thermometer
5. Smallest division = 1 mm. Least count =  
A. 1 cm B. 10 mm C. 1 mm D. 0.01 mm
6. Which is greatest?  
A. 150 cm B. 1 m C. 1.2 m D. 100 cm

**B. Application-Based Questions**

7. A ruler has a damaged zero mark. A pencil starts at 1.5 cm and ends at 16.2 cm. Find its length.
8. Which is the most reasonable measurement for the thickness of a coin? A. 2 km B. 2 m C. 2 cm D. 2 mm
9. A classroom is stated as 800 cm long by one student and 8 m long by another. Are the measurements different? Explain.
10. One bottle has a mass of 750 g and another has a mass of 1 kg. Which is heavier and by how much?

**C. Numerical Questions**

11. Convert: (a) 4.5 km into metres (b) 3250 m into km (c) 6.25 m into cm (d) 4800 g into kg.
12. A race lasts 2 minutes 35 seconds. Express the total time in seconds.
13. A rope is 3 m 45 cm long. Express its length completely in centimetres.
14. A weighing balance shows 2 kg 350 g. Express the mass in grams.

**D. Homi Bhabha Challenge Questions**

15. Student A measures a pencil as 14.8 cm; Student B measures it as 148 mm. Are the measurements different? Explain.
16. A ruler's zero mark is broken. A pencil gives readings of 2.3 cm and 13.7 cm. Find its actual length.
17. Choose the best instrument for: (a) curved road length (b) mass of vegetables (c) sprinter's time (d) water temperature.
18. Arrange in increasing order: 2 m, 150 cm, 2500 mm, 1.8 m.
19. A student converts 5 kg 250 g into grams and gets 525 g. Identify the mistake and give the correct answer.
20. One instrument has least count 1 cm and another 1 mm. Which is more precise? Give a reason.

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**ANSWER KEY & SOLUTIONS**

- |                                                                                              |
|----------------------------------------------------------------------------------------------|
| 1. C. m                                                                                      |
| 2. D. km                                                                                     |
| 3. $3.5 \times 100 = 350$ cm (B)                                                             |
| 4. B. Stopwatch                                                                              |
| 5. C. 1 mm                                                                                   |
| 6. $1.2$ m = 120 cm, so 150 cm (A)                                                           |
| 7. $16.2 - 1.5 = 14.7$ cm.                                                                   |
| 8. D. 2 mm.                                                                                  |
| 9. No. 800 cm = 8 m; same measurement.                                                       |
| 10. 1 kg = 1000 g; difference = 250 g. The 1 kg bottle is heavier.                           |
| 11. (a) 4500 m; (b) 3.25 km; (c) 625 cm; (d) 4.8 kg.                                         |
| 12. $2 \times 60 + 35 = 155$ s.                                                              |
| 13. $300 + 45 = 345$ cm.                                                                     |
| 14. $2000 + 350 = 2350$ g.                                                                   |
| 15. 14.8 cm = 148 mm, so both are equal.                                                     |
| 16. $13.7 - 2.3 = 11.4$ cm.                                                                  |
| 17. (a) Measuring tape (b) weighing balance (c) stopwatch (d) thermometer.                   |
| 18. $150 \text{ cm} < 1.8 \text{ m} < 2 \text{ m} < 2500 \text{ mm}$ .                       |
| 19. 5 kg = 5000 g; $5000 + 250 = 5250$ g.                                                    |
| 20. The 1 mm least-count instrument is more precise because it measures smaller differences. |